

Inference of Pairwise Interactions from Strain Frequency Data Across Settings and Context-Dependent Mutual Invasibilities

Thi Minh Thao Le

Instituto Superior Tecnico, University of Lisbon
Lisbon, Portugal

- 1 Introduction & overview
- 2 Parameter estimation progress
- 3 Results
- 4 Discussion

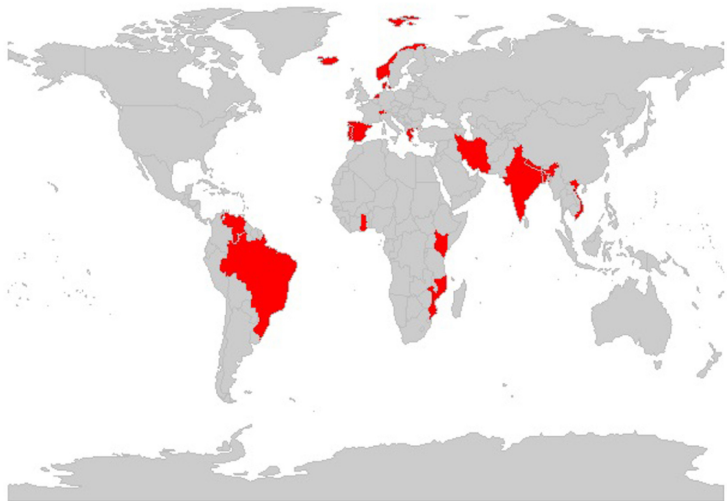


Figure 1: Country map. Map of the geographical locations from around the world, where the cross-sectional studies reported single- and co-colonization prevalences of ***Streptococcus pneumoniae*** carriage in predominantly healthy children.

Why study pneumococcal strain interactions?

Streptococcus pneumoniae (the pneumococcus)

colonizes the nasopharynx of healthy children worldwide. It is a leading cause of **pneumonia**, **bacterial meningitis**, and **otitis media** (middle ear infection), and remains one of the top infectious killers of children under 5 globally.

- **Extraordinary diversity:** Over **92 serotypes** are distinguished by their polysaccharide capsule. Multiple serotypes circulate simultaneously in every population, and a single child can carry more than one serotype at the same time (**co-colonization**).
- **Strains interact during colonization:** When two serotypes share the same host, one may suppress or facilitate the other. These pairwise interactions shape which serotypes dominate and which remain rare.
- **Vaccine design challenge:** Current conjugate vaccines target a subset of serotypes. Eliminating vaccine serotypes can open ecological niches for non-vaccine serotypes (**serotype replacement**). Predicting replacement requires understanding the interaction network among all circulating strains.

Multi-strain coinfection SIS system

Allow a host already colonized by strain i to acquire strain j , creating co-colonized compartments I_{ij} .

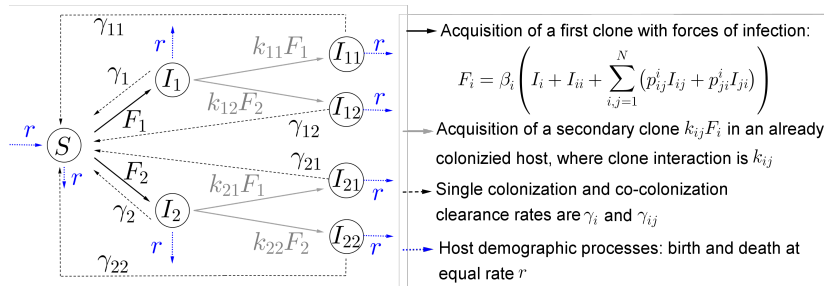


Figure 2: Model diagram when $N = 2$. This introduces new families of parameters: co-colonization susceptibilities k_{ij} , transmission probabilities from mixed hosts p_{ij}^s , and co-infection clearance rates γ_{ij} .

The challenge: With N strains, the full system has $O(N^2)$ equations. Direct analysis is intractable.

What do the raw data look like?

Each cross-sectional study swabs a sample of n children and classifies every child into one of three states:

State	Count	Proportion
Susceptible (no pneumococcus)	n_S	$S = n_S/n$
Singly colonized (one serotype)	n_I	$I = n_I/n$
Co-colonized (two+ serotypes)	n_D	$D = n_D/n$
Total	n	$S + I + D = 1$

- Total carriage prevalence: $T = I + D = 1 - S$.
- These three proportions (S, I, D) are the **only inputs** needed to estimate all model parameters.
- Example (Iran, $n = 1291$): $S = 65.8\%$, $I = 16.5\%$, $D = 17.7\%$.

Source [2] Dekaj & Gjini, *Theor. Popul. Biol.* 156 (2024).

From proportions to parameters

At **endemic equilibrium**:

$$S^* = \frac{1}{R_0}, \quad I^* = \frac{R_0 - 1}{R_0[1 + (R_0 - 1)k]}, \quad D^* = \frac{k(R_0 - 1)^2}{R_0[1 + (R_0 - 1)k]}.$$

Setting observed proportions equal to model predictions and solving gives

closed-form estimators:

Basic reproduction number

$$\hat{R}_0 = \frac{1}{1 - T} \quad (\text{from total carriage prevalence } T = I + D)$$

Mean susceptibility to co-colonization

$$\hat{k} = \frac{D}{I} \cdot \frac{1}{\hat{R}_0 - 1} = \frac{D}{I} \cdot \frac{1 - T}{T}$$

Single-to-co-colonization ratio

$$\hat{\mu} = \frac{I}{D} \quad \left(\text{equivalently } \mu = \frac{1}{(\hat{R}_0 - 1)\hat{k}} \right)$$

Worked example & uncertainty

Iran

($n = 1291$): $S = 0.658$, $I = 0.165$, $D = 0.177$.

$$\textcircled{1} \quad T = I + D = 0.342, \text{ so } \hat{R}_0 = \frac{1}{1 - 0.342} = 1.52.$$

$$\textcircled{2} \quad \hat{k} = \frac{D}{I} \cdot \frac{1}{\hat{R}_0 - 1} = \frac{0.177}{0.165} \cdot \frac{1}{0.52} = 2.06.$$

$$\textcircled{3} \quad \hat{\mu} = \frac{I}{D} = \frac{0.165}{0.177} = 0.93.$$

Interpretation: Iran has low transmission ($\hat{R}_0 \approx 1.5$) but high co-colonization susceptibility ($\hat{k} > 1$, facilitation). The ratio $\hat{\mu} < 1$ means co-colonization is *more common* than single colonization.

95% confidence intervals: obtained by multinomial resampling (500 draws of (n_S, n_I, n_D) from the observed proportions and sample size n), re-estimating $\hat{R}_0$ and $\hat{k}$ each time, and taking the 2.5% and 97.5% quantiles.

Motivation

- **The puzzle:** Different countries report strikingly different serotype compositions. Are these differences driven by different strain–strain interactions, or by the *same* interactions playing out under different epidemiological conditions?
- **The scale of the problem:** 92 serotypes imply $92 \times 91 = 8,372$ directed pairwise interactions to estimate — a massive inference challenge.
- **The data limitation:** Most available data are cross-sectional “snapshots” of serotype frequencies (swab children, identify strains, report proportions). We observe the *outcome* of competition, not the competition itself. From a single snapshot, many interaction matrices could produce the same frequencies — the inverse problem is ill-posed.
- **Our strategy:** Combine snapshots from **5 sites** with different ecological contexts. Each site provides a different “view” of the same underlying interactions, progressively constraining the solution.

Modeling framework

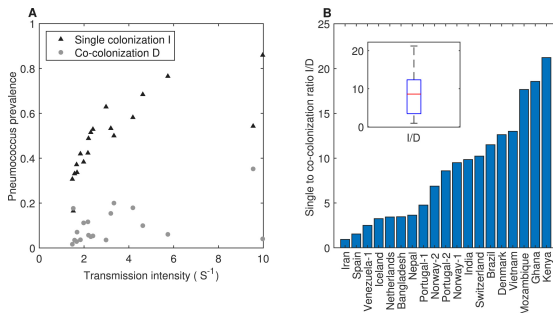


Figure 3: Visualization and summary of the selected epidemiological data for this study

- **Key Concept:** Starting from **SIS model with co-infection** as the foundation. Using **replicator dynamics** to link epidemiological cross-sectional data with mechanistic strain interactions.
- **Parameter K_{ij} in SIS model:** The susceptibility of a host already colonized by strain i to be co-colonized by strain j . $K_{ij} > 1$: facilitation; $K_{ij} < 1$: competition.
- **Similarity assumption:** $K_{ij} = k + \epsilon\alpha_{ij}$, $0 < \epsilon \ll 1$.

Id	Location	Single to co-colonization ratio (I/D) $\hat{\mu}$ (95% CI)	Basic reproduction number $\hat{R}_0$ (95% CI)	Mean susceptibility to co-colonization $\hat{k}$ (95% CI)
1	Iran	0.93 (0.66–1.06)	1.52 (1.5–1.58)	2.06 (1.9–2.6)
2	Spain	1.29 (0.27–2.27)	9.55 (7.5–21)	0.09 (0.07–0.18)
3	Venezuela-1	2.5 (0.51–4.3)	3.33 (2.94–5.56)	0.17 (0.12–0.43)
4	Iceland	3.25 (2.03–3.86)	4.18 (4.02–4.87)	0.1 (0.09–0.13)
5	Netherlands	3.46 (2.45–3.88)	3.19 (3.1–3.5)	0.13 (0.12–0.16)
6	Bangladesh	3.63 (1.62–4.69)	2.44 (2.33–2.9)	0.19 (0.16–0.32)
7	Nepal	3.95 (2.49–4.71)	1.98 (1.92–2.14)	0.26 (0.23–0.35)
8	Portugal-1	4.77 (2.33–6.23)	2.78 (2.62–3.31)	0.12 (0.1–0.19)
9	Norway-2	5.71 (3.31–7.03)	5.06 (4.78–6.11)	0.04 (0.04–0.06)
10	Portugal-2	6.42 (3.56–7.8)	3.45 (3.3–4.01)	0.06 (0.06–0.09)
11	Norway –1	6.92 (4.12–8.25)	4.49 (4.27–5.27)	0.04 (0.04–0.06)
12	India	8.59 (4.95–10.85)	2.2 (2.13–2.43)	0.1 (0.08–0.14)
13	Switzerland	9.57 (4.37–12.86)	2.06 (1.98–2.35)	0.1 (0.08–0.17)
14	Brazil	9.85 (4.09–13.93)	2.4 (2.28–2.86)	0.07 (0.06–0.13)
15	Denmark	10.23 (5.44–13.16)	2.3 (2.22–2.56)	0.08 (0.06–0.12)
16	Vietnam	13 (5.98–18.24)	1.67 (1.62–1.8)	0.12 (0.09–0.21)
17	Mozambique	16.8 (10.05–20.38)	6.44 (6.14–7.69)	0.01 (0.009–0.015)
18	Ghana	18.57 (10.35–24.34)	1.48 (1.45–1.55)	0.11 (0.09–0.18)
19	Kenya	21.34 (12.64–26.2)	10.45 (9.79–12.35)	0.00496 (0.004–0.007)
Global mean		7.94 (4.27–10.21)	3.74 (3.47–4.94)	0.2 (0.18–0.29)
Global mode		3.55 (2.47–4.7)	2.42 (2.3–2.4)	0.1 (0.08–0.18)

Figure 4: Estimated epidemiological parameters for each *Streptococcus pneumoniae* cross-sectional carriage study in young children, across 19 epidemiological settings around the world, using the SIS model

Model and properties

Replicator equation: (for computational efficiency & interaction between strains)

$$\frac{d}{d\tau} z_i = \Theta z_i \left(\sum_{j \neq i} \lambda_i^j z_j - \sum_{k \neq j} \lambda_j^k z_j z_k \right).$$

- $\bar{\lambda} = Q = \sum_{k \neq j} \lambda_j^k z_j z_k$: mean invasion fitness of the system.
- Pairwise invasion fitness: $\lambda_i^j = \alpha_{ji} - \alpha_{ij} + \mu (\alpha_{ji} - \alpha_{ij})$.
- **Context Parameter** $\mu = I^*/D^* = 1/k(R_0 - 1)$: ratio of single to co-colonization prevalence, acts as a "resource gradient."

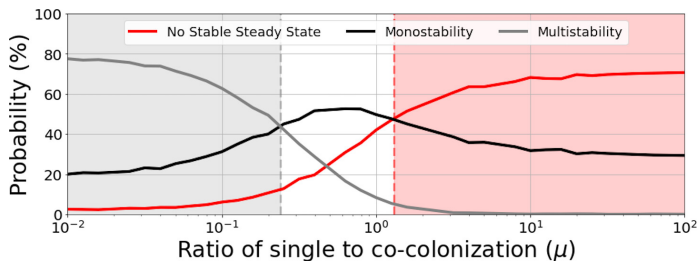


Figure 5

Rationale for Site Selection

- **Broad Variation in μ :** The five geographical environments span a wide range of single-to-coinfection ratios (μ), from 0.93 (Iran) to 16.8 (Mozambique).
- **Testing Multiple Dynamical Hypotheses:** The divergence in μ allows for the evaluation of two distinct mathematical assumptions within the replicator equation:
 - $\mu > 10$ (Brazil, Denmark, Mozambique): Models non-equilibrium dynamics (snapshots).
 - Low μ (Nepal, Iran): Tests the stable steady-state assumption.
- **Conserved Microbial Network Structure:** Highly similar rank order abundance distributions. A shared core networks of mutual serotypes facilitates the iterative cross-country estimation of the global α -matrix.

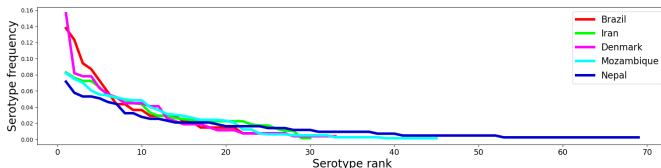


Figure 6: Rank-order-abundance distributions

Data overview: Five sites

Table 1: *S. pneumoniae* in five eco-epidemiological contexts with different global colonization and co-colonization parameters. Source: [1], [3], [4], [5], [6]

<i>Study site</i> (Nr.serotypes/ Sample size)	<i>Single to cocolo- nization ratio</i> $\hat{\mu}$ (95% CI)	<i>Basic reproduc- tion number</i> $\hat{R}_0$ (95% CI)	<i>Mean susceptibility to co-colonization</i> $\hat{k}$ (95% CI)
Brazil (26/242)	9.85 (4.09-13.93)	2.4 (2.28-2.86)	0.07 (0.06-0.13)
Iran (30/1291)	0.93 (0.66-1.06)	1.52 (1.5-1.58)	2.06 (1.9-2.6)
Denmark (33/437)	10.23 (5.44-13.16)	2.3 (2.22-2.56)	0.08 (0.06-0.12)
Mozambique (45/927)	16.8 (10.05-20.38)	6.44 (6.14-7.69)	0.01 (0.009-0.015)
Nepal (69/600)	3.95 (2.49-4.71)	1.98 (1.92-2.14)	0.26 (0.23-0.35)

- Snapshot of dynamics: Brazil, Denmark, Mozambique (Nepal)
- Stable equilibrium: Iran, Nepal

- 1 Introduction & overview
- 2 Parameter estimation progress**
- 3 Results
- 4 Discussion

- **Order 1 (Incremental):** Brazil → Denmark → Mozambique → Iran → Nepal (Based on increasing n).
- **Order 2 (Stability):** Nepal → Brazil → Denmark → Mozambique → Iran.
- **Logic:** Nepal provides the most “degrees of freedom” (69 strains) to fill the 92×92 interaction matrix α early (56.25% in step 1).

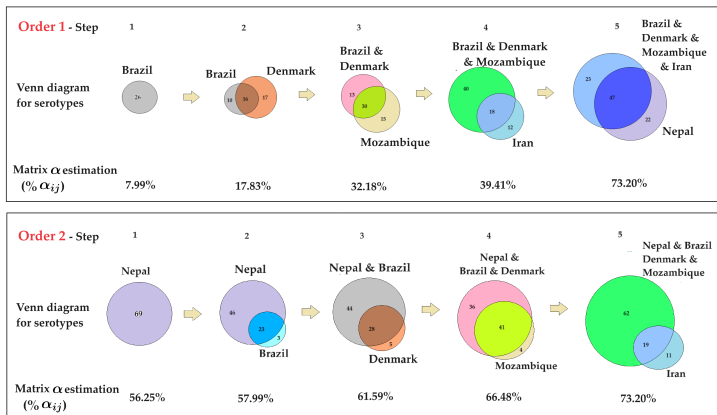


Figure 7: Venn diagram representation of parameter estimation process.

Why Particle Swarm Optimization (PSO)?

Particle Swarm Optimization (PSO)

A swarm of candidate solutions (“particles”) explores the search space simultaneously. Each particle adjusts its trajectory based on its own best-known position and the swarm’s global best, balancing **exploration** (searching new regions) with **exploitation** (refining promising ones).

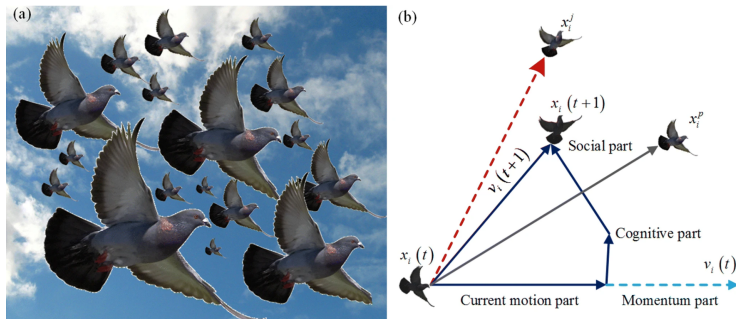


Figure 8: PSO algorithm. **a.** The movement of particle in the environment; **b.** Velocity and position updates of particles in the PSO algorithm

Why not classical alternatives?

- **Gradient-based methods** (e.g. L-BFGS, Newton): require smooth, differentiable objectives. Our landscape involves ODE simulations, eigenvalue constraints, and a non-convex objective $\rightarrow$ gradients are unavailable or unreliable.
- **Grid/random search**: with $\alpha_{ij} \in [-10, 10]$ and thousands of parameters, exhaustive search is computationally impossible.

Why PSO fits this problem?

- **Gradient-free**: only evaluates the objective function (RMSE), never needs derivatives.
- **Handles high dimensionality**: scales to the large number of α_{ij} entries estimated at each step.
- **Constraint-friendly**: stability constraints (negative eigenvalues of the Jacobian) are incorporated by penalizing infeasible particles rather than reformulating the problem.
- **Parallelizable**: particles are evaluated independently, enabling straightforward parallelization across CPU cores.

Optimization & Numerical Walk-through

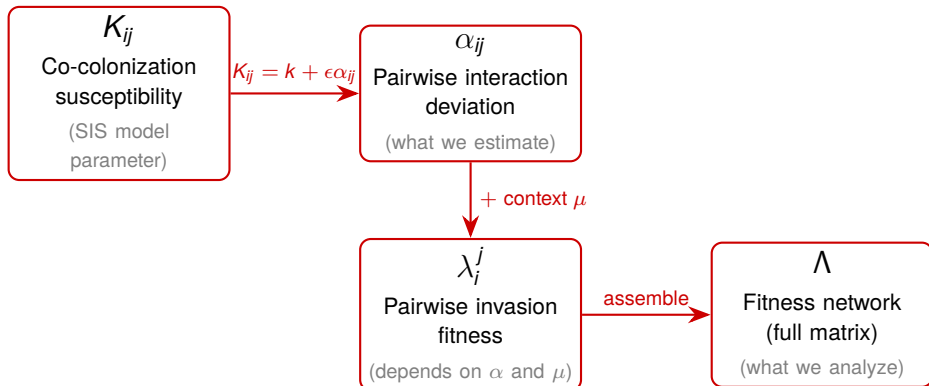
Goal

Minimize error between model prediction and observed frequencies.

Algorithm

- ➊ **Input:** Observed frequencies z^{data} and site μ .
- ➋ **Mapping:** For a proposed α_{ij} matrix (search space: $\alpha_{ij} \in [-10, 10]$), calculate the Λ matrix using the λ_i^j formula.
- ➌ **Simulation:** For Dyn sites, solve the ODE until $\tau = 50$. For Steady State (SS) sites, compute the interior equilibrium.
- ➍ **Objective Function:** $\text{RMSE} = \frac{1}{\sqrt{n}} \sqrt{\sum_{i=1}^n |z_i^{\text{model}} - z_i^{\text{data}}|^2}$. Constraint: For SS, the Jacobian at z^* must have all eigenvalues with strictly negative real parts ($\text{Re } \lambda < 0$ for all $\lambda \in \sigma(J)$).
- ➎ **Iteration:** PSO updates particle positions to minimize RMSE (for SS sites, keep satisfying stability).

From biology to fitness: notation summary



Key point: α is **universal** across sites; μ **varies** by site;
therefore Λ is **context-dependent**.

- 1 Introduction & overview
- 2 Parameter estimation progress
- 3 Results**
- 4 Discussion

Quality of Fit

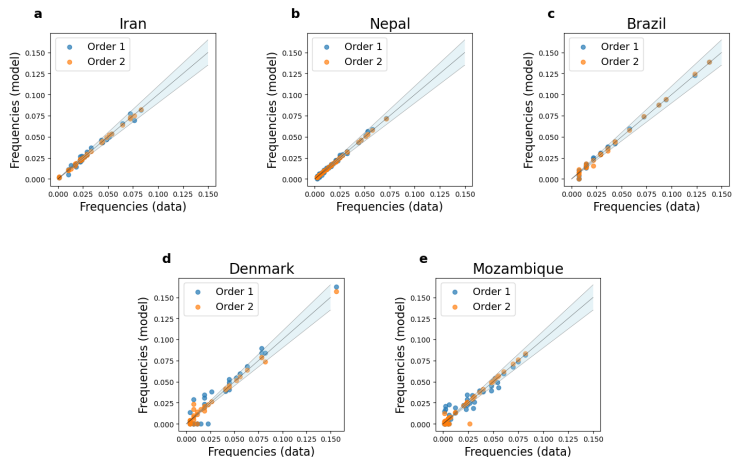


Figure 9: The mean square errors of 5 sites Iran, Nepal, Brazil, Denmark and Mozambique, respectively are: **Order 1** (0.0027, 0.0012, 0.003, 0.0094, 0.0065) and **Order 2** (0.0007, 0.000019, 0.0024, 0.0046, 0.0047).

Statistical Properties of Estimated α_{ij}

- The estimated global interaction matrix (73.2% filled) is heavy-tailed. **Order 1** distribution: Mean ≈ 0.0038 , SD ≈ 1.16 . **Order 2** distribution: Highly peaked at 0 with negative outliers (near -10).
- Technical insight: Negative diagonal values α_{ii} in Nepal (Order 2) reflect stabilizing mechanisms required for large-scale coexistence.

Invasion Resistance Q (system's resistance to new strain entry)

- **Observation:** Under stable coexistence (Order 2), Q values are significantly higher (all > 5) compared to Order 1 (all < 1).
- **Insight:** A mature, stable ecosystem is mathematically harder to invade.

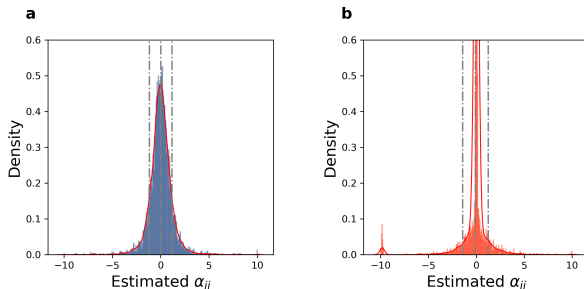


Figure 10: α matrix distribution

Emergent Mutual Invasion Fitness Networks

- **Order 1:** Dominated by $+/-$ edges (exclusion or predator-prey outcomes) $\rightarrow$ complex oscillatory dynamics in simulations. **Order 2:** Dominated by $+/+$ edges (mutual coexistence), favoring stable SS.

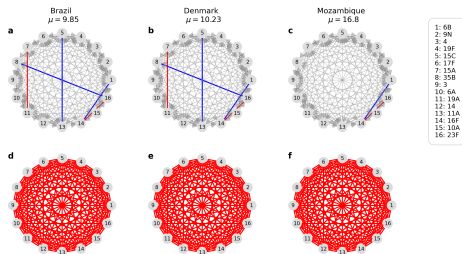


Figure 11: Emergent invasion fitness network

- **Key finding:** Imposing stability for 69 strains in Nepal forces the global interaction matrix toward a coexistence regime.

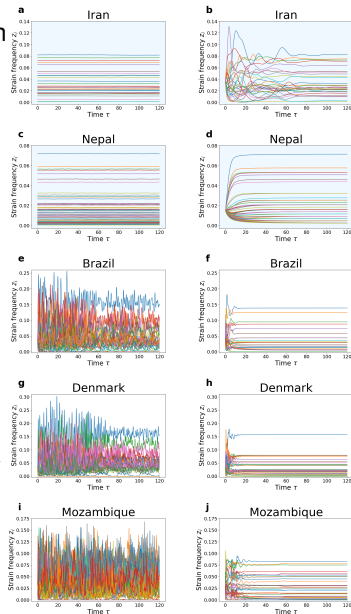


Figure 12: Model predictions

Symmetry Analysis of α_{ij} -matrix vs. Λ

Symmetry Index (s) of a matrix $M = (m_{ij})_{1 \leq i, j \leq N}$

$$s \equiv \frac{\langle m_{ij} m_{ji} \rangle}{\langle m_{ij}^2 \rangle} = 2 \left(\sum_{i=1}^N \sum_{j>i}^N m_{ij} m_{ji} \right) / \left(\sum_{i=1}^N \sum_{j \neq i}^N m_{ij}^2 \right).$$

$s \approx 1$: symmetric (Mutualistic/Competitive), $s \approx 0$: random, $s \approx -1$: anti-symmetric (Predator-Prey/Oscillatory).

Key Findings for Interaction Matrix (α)

- In both Order 1 and Order 2, the α matrix exhibits a symmetry index near 0.
- The underlying co-colonization vulnerabilities are randomly distributed.

Key Findings for Fitness Matrix (Λ)

- **Order 1:** Λ is nearly anti-symmetric ($s \approx -1$) for high- μ sites (except Iran).
- **Order 2:** Λ shifts toward random or symmetric values (Nepal, Iran), consistent with stable coexistence.

Mathematical Implication The global context μ "tunes" a random biological backbone (α) into either a stable or an oscillatory functional network (Λ).

- 1 Introduction & overview
- 2 Parameter estimation progress
- 3 Results
- 4 Discussion**

Computational Performance & Algorithm Limitations

- **Execution time analysis:**

- Order 1 (incremental n): Required approximately **84–96 hours**.
- Order 2 (stability first): Significantly more efficient, requiring only **30–36 hours**.

- **The Nepal bottleneck:**

- Computing stability for Nepal ($n = 69$) is the most time-consuming phase.
- Evaluating the Jacobian of a 69×69 matrix to ensure all eigenvalues have negative real parts is computationally intensive.

- **Sequential vs. simultaneous estimation:**

- Simultaneous estimation of the 92×92 matrix is currently unfeasible with standard resources.
- Sequential fitting reduces the degrees of freedom in later steps, as many α_{ij} values are fixed by preceding sites.

- **PSO disadvantages:**

- Exponential search space: As dimensionality increases, the search space grows exponentially, hindering efficient navigation.
- Stochasticity: Random components in velocity updates mean that convergence to a global optimum is not guaranteed in every run.

What if context doesn't matter?

Alternative hypothesis (Stress-Gradient Hypothesis)

The single-to-co-colonization ratio μ is **the same everywhere**. If true, the pairwise invasion fitness λ_i^j would be **identical across all five sites** — one universal fitness network would govern all populations.

Setup of the test:

- Fix $\mu \approx 6$ globally (average across sites).
- Estimate a single α matrix that must simultaneously reproduce the observed serotype frequencies in Brazil, Iran, Denmark, Mozambique, and Nepal.
- Compare fit quality and network structure against the context-dependent model (site-specific μ).

Prediction: If the SGH is correct, fit quality should be comparable and the resulting Λ network should be consistent across sites.

Evidence: context-dependence cannot be removed

The data reject the universal- μ model on three levels:

- ❶ **Worse fit:** RMSE increases significantly compared to the site-specific μ model. The same interaction matrix cannot reproduce the different frequency profiles observed across sites.
- ❷ **Fitting impasse:** For later countries in the nesting sequence, it becomes nearly impossible to reduce the error below 1.5%. The constraints imposed by earlier sites actively conflict with the data from later ones.
- ❸ **Wrong network structure:** A uniform high μ forces the fitness matrix Λ to be nearly **anti-symmetric** ($s \approx -0.98$) across all sites. This produces a “predator–prey” network in which:
 - stable coexistence becomes **rare**,
 - oscillatory dynamics become **the rule**.

This contradicts the observed long-term coexistence of dozens of serotypes in every population.

Verdict

Context-dependent fitness (μ varying by site) is not just a better fit — it is **necessary** to produce biologically plausible interaction networks.

Summary & Future Outlook

- **Mechanism of coexistence:** The study identifies a common ecological underpinning in which a "universal" biological backbone α is modulated by a local "resource gradient" μ .
- **Stability as an organizing principle:**
 - Imposing stability constraints (Order 2) revealed a coexistence regime dominated by positive (+/+) fitness edges, which provides higher invasion resistance (Q).
- **Trait-based paradigms:**
 - Future work aims to move from a "serotype predicts prevalence" framework to a "serotype predicts interactions predicts prevalence" paradigm.
 - There is potential to link estimated α_{ij} values to genetic or antigenic distances between strains.
- **Methodological Scalability:** This framework serves as a proof-of-concept for quantifying high-dimensional interactions in other biodiverse ecosystems, such as the human microbiota.
- **Final Conclusion:** Context-dependent fitness is essential to mechanistically link diverse cross-sectional data across varying epidemiological settings.

References I



Tolulope Adebajo, Fernanda C Lessa, Helio Mucavele, Benild Moiane, Alberto Chauque, Fabiana Pimenta, Sergio Massora, Maria da Gloria Carvalho, Cynthia G Whitney, and Betuel Sigauque.

Pneumococcal carriage and serotype distribution among children with and without pneumonia in mozambique, 2014-2016.

PloS one, 13(6):e0199363, 2018.



Ermanda Dekaj and Erida Gjini.

Pneumococcus and the stress-gradient hypothesis: A trade-off links r_0 and susceptibility to co-colonization across countries.

Theoretical Population Biology, 156:77–92, 2024.



Zitta B Harboe, Hans-Christian Slotved, Helle B Konradsen, and Margit S Kaltoft.

A pneumococcal carriage study in danish pre-school children before the introduction of pneumococcal conjugate vaccination.

The open microbiology journal, 6:40, 2012.

References II



Rama Kandasamy, Meeru Gurung, Anushil Thapa, Susan Ndimah, Neelam Adhikari, David R Murdoch, Dominic F Kelly, Denise E Waldron, Katherine A Gould, Stephen Thorson, et al.

Multi-serotype pneumococcal nasopharyngeal carriage prevalence in vaccine naive nepalese children, assessed using molecular serotyping.

PloS one, 10(2):e0114286, 2015.



HG Rodrigues, TCA Pinto, RR Barros, LM Teixeira, and FPG Neves.

Pneumococcal nasopharyngeal carriage among children in brazil prior to the introduction of the 10-valent conjugate vaccine: a culture-and pcr-based survey.

Epidemiology & Infection, 145(8):1720–1726, 2017.



S Rafiei Tabatabaei, F Fallah, F Shiva, AR Shamshiri, M Hajia, M Navidinia, A Karimi, and M Rahbar.

Multiplex pcr assay for detection of pneumococcal serotypes in nasopharyngeal samples of healthy children; tehran, 2009-2010.

Annual Research & Review in Biology, pages 3780–3790, 2014.